Sprint 1 – Manager with Reserve

Student Information

Integrity Policy: All university integrity and class syllabus policies have been followed. I have neither given, nor received, nor have I tolerated others' use of unauthorized aid.

I understand and followed these policies: (•) Yes ( ) No

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Date: 1/31/2024

Submission Details

Final ***Changelist*** number: 205271

Verified build: (•) Yes ( ) No

Required Configurations: DEBUG only

YouTubeLink: https://www.youtube.com/watch?v=W8vrGxoB9m4

Discussion (What did you learn):

Creating a mangaer with reserve using linked list

Verify Builds

- Follow the Piazza procedure on submission
  - Verify your submission compiles and works at the changelist number.
- Verify that only MINIMUM files are submitted
  - No – Generated files
    - *.pdb, *.suo, *.sdf, *.user, *.obj, *.exe, *.log, *.pdb, *.db, *.user
    - Anything that is generated by the compiler should not be included
  - No – Generated directories
    - /Debug, /Release, /Log, /ipch, /.vs
- Typical files project files that are required
  - *.sln, *.csproj, *.cs,
  - App.config, AssemblyInfo.cs, CleanMe.bat
  - Resources Directory:
    - *.tga, *.dll, *.wav, *.glsl, *.azul

Standard Rules

Submit multiple times to Perforce

- Submit your work as you go to perforce several times
  - Design Patterns – no minimum
  - Sprints – minimum of 5 real submissions (generally 10+)
    - As soon as you get something working, submit to perforce
    - Have reasonable check-in comments
      - Points will be deducted if minimum is not reached

Submission Report

- Fill out the submission Report
  - No report, no grade

Code and project needs to compile and run

- Make sure that your program compiles and runs
  - Warning level 4
  - NO Warnings or ERRORS
    - Your code should be squeaky clean.
  - Code needs to work “as-is”.
    - No modifications to files or deleting files necessary to compile or run.
  - All your code must compile from perforce with no modifications.
    - Otherwise it's a 0, no exceptions

Project needs to run to completion

- If it crashes for any reason…
  - It will not be graded and you get a 0

No Containers

- Containers (No automatic containers or arrays
- Template or generic parameters
- No arrays
  - You need to do this the old fashion way - ***YOU EARNED IT***

Leave Project Settings

- Do NOT change the project or warning level
  - Any changing of level or suppression of warnings is an integrity issue

Simple C#

- No .Net
- We are using the basics
  - Types:
    - Class, Structs, intrinsic types (int, float, bool, etc…)
    - NO arrays allowed!
  - Basics language features
    - Inheritance, methods, abstract, virtual, etc…

No Debug code or files disabled

- Make sure the program has only active code
  - If you added debug code or commented out code,
    - please return to code to active state or remove it

Adding files to this project

- Make sure you add the files in the appropriate sub-directories
- Make sure any new files are successfully integrated into the project
- Make sure your new files are submitted to Perforce

Due Dates

- See Piazza for due date and time
- Submit program perforce in your student directory assignment supplied.
- Fill out your this ***Submission Report*** and commit to perforce
  - ***ONLY*** use Adobe Reader to fill out form, all others will be rejected.
  - Fill out the form and discussion for full credit.

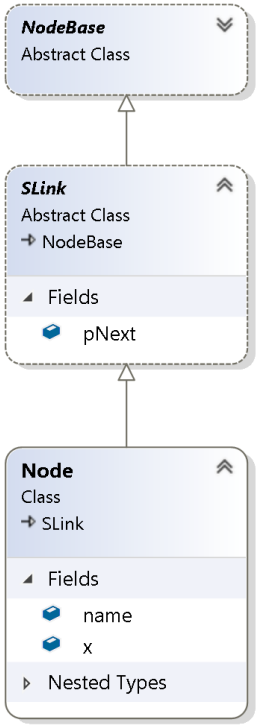
Goals

- Manager with Reserve using Single Linked List
  - A small twist on the demos from class
- Create a reserve manager using single linked lists
  - Add, Remove, Find methods
  - SLink manager is internally created inside the Manager.

Assignments

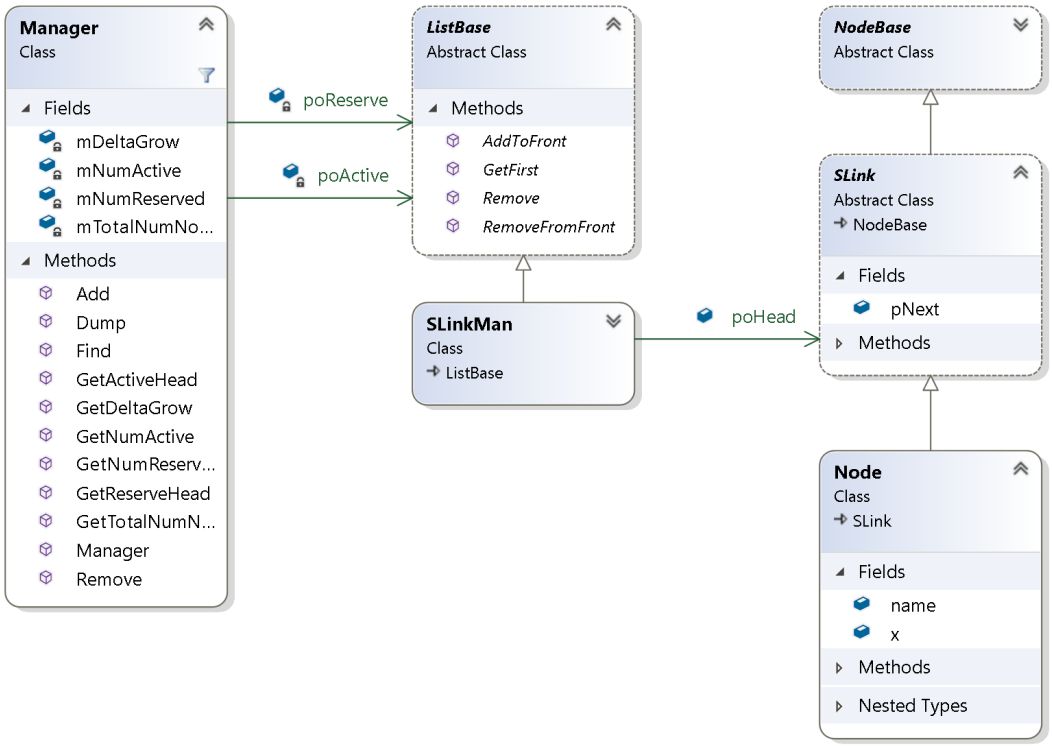
Grading

- Game Engine has a unit test in Main
- Make sure you pass all 4 tests.
- Code a reserve manager using the provided class, Node:



- Feel Free to add classes to the project
  - Modify the Class Diagram
  - Manager class is stubbed out
  - Please fill in the missing methods
  - Feel free to add more method and data

- Here is my UML diagram to give you some ideas:



- Do NOT modify the unit tests
  - Conform to the methods provided.
- You will need to add files to the project
  - My suggestion
    - Always create the directories and files in the tool
    - This way they are in the correct location
    - Then cut and paste into those files
- Namespace
  - Keep every file using namespace SpaceInvaders
- Demonstrate you have all the tools and setup working

Required Features

Manager

- Create a Manager
  - Manager actually has 2 linked list attached an Active List and a Reserved List
    - essentially 2 head pointers
  - All nodes are ***single linked lists***
- Manager(…) - Initialized you manager with a reserved size and growth size
  - Example, say you initialize the reserve size with 5 nodes, growth size is 2 nodes
    - You will create on construction of the manager
      - 5 linked nodes attached to the reserved list.

- You set the growth size member variable to 2
- The Active list will have nothing attached only null pointer.

- Generally
  - Adding a node
    - You move a node from the reserved list and add it to the active list
      - Insert the node to the front of the list
        - In front for speed…. Think about it O(1)
    - If the reserved list is empty
      - Refill the reserve list with the growth size
      - The reserved list will be filled with that many nodes, then you can move one from the reserve and transfer it to the active list
        - (might seem unnecessary to move to reserve then immediately transfer to active… but it reduces complexity)
  - Remove a node
    - Remove any given node by address
      - Worse case O(N) – it's a single linked list
    - Add it back to the Reserve List
      - To the front of the list O(1) complexity
- Add(...) - creates and return node
  - moves one from the reserved list and inserts the node to the active list
- Remove(...) - recycles the node
  - removes the node from active list and transfers it to the reserved list
- Ability to report stats – data we can use
  - number of active, reserved, growth number, total number, etc...
- Find(...) - searches through active list and returns node
  - Search by Name
    - Hint – you'll need to have casting to make this possible

Node

- Holds data (int x)
- Holds name (enumeration)
- Inherits from SLink (single linked list)

SLink

- Holds the next references
- Node derives from SLink
- Used for implementing in terms of... paradigm

NodeBase

- Abstract base class (place holder)
- Do nothing in this class

Video Capture

- Have Visual Studio open and ready to go.
- Start recording
    - Show compiling and output window running the tests (All 4 test passing)
- Commentary:
    - Speak clearly and loud
    - Show off a section of the code (suggestion the Find() function)
        - Show the UML diagram (in the tool)
        - Show your code an quickly explain it.
    - Duration
        - Approx. 2 min long YouTube video capture
            - Do not exceed 5 min
        - Use Zoom recording
            - High resolution
            - Good sound quality
            - Show on the capture… it compiling and then running
        - Place YouTube link in PDF
        - Make sure it public and runs without passwords or login
        - Watch your video
            - Can you see the code clearly
            - Can you hear your voice clearly
- Don't lose points
    - Make sure everything is submitted properly
    - Don't record beyond 5 min.
    - You audio volume was too quiet (test it and verify it)
    - Your YouTube link needs to work "As-IS" with no login
        - Don't make it private… make it unlisted
    - You check-in too many files or temporary files
        - Make sure you run CleanMe.bat before perforce submission

General guidelines:

- Post on Piazza questions and clarifications
- Make sure you delete these using directives (we are not using them)
    - `using System.Collections.Generic;`
    - `using System.Linq;`
    - `using System.Text;`
    - `using System.Threading.Tasks;`

Development

- Use the project in Sprint1
- Do not copy temporary files or submit extra data
- Make sure your run CleanMe.bat before adding to perforce

Submission

- Submit your SprintX directory into perforce:
  - /student/<yourname>/SprintX/...
    - You need to submit a complete C# project
  - Solution, project and C# files (whatever it takes to build the project)
    - Do not submit anything that is auto generated
  - Run the supplied CleanMe.bat before submission
    - Should cleanup files
- Fill out the Submission report and submit that pdf to your student directory

Validation

Simple checklist to make sure that everything is submitted correctly

- Is the project compiling and running without any errors or warnings?
- Does the project runs ***ALL*** without crashing?
- Is the submission report filled in and submitted to perforce?
- Follow the verification process for perforce
  - Is all the code there and compiles “as-is”?
  - No extra files
- Submit the YouTube link to PDF

Hints

Most assignments will have hints in a section like this.

- Reuse the work you did in the Linked List assignment
- Make sure you are using Single Linked List
- Study the projects in class lecture

Troubleshooting

- Print to the output window to help you debug
  - It really helps
- Ask for clarification on piazza